Revamp Investor CMS Technical Documentation

Dokumentasi teknis komprehensif mengenai arsitektur sistem, komparasi mendalam seluruh query database antara Smarty CMS dan Revamp CMS (CodeIgniter 4), spesifikasi seluruh pemanggilan API, panduan pra-produksi (*pre-production checklist*), serta catatan integrasi frontend.

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1. Ringkasan Arsitektur & Perubahan Sistem

1.1 Perbandingan Arsitektur

Karakteristik	Smarty CMS	Revamp CMS
Framework & Versi	PHP 5.6 Prosedural Native + Smarty 2/3	CodeIgniter v4.7.4 (PHP 8.3.28 / min. PHP ^8.2, Strict Types, PSR-4)
Pola Routing	Berbasis file fisik (`/administration/pages/article/index.php`)	Centralized Router (`app/Config/Routes.php`) dengan Route Guard & Filter
Interaksi Database	Prosedural `adodb` / raw string MySQL query concats	CodeIgniter 4 Query Builder & Model Layer
Konvensi Case-Sensitivity	Raw SQL dengan string literal (`tblArticle` , `tblCategory`)	Model & Query Builder CamelCase (`Article` , `Category`), otomatis di-prefix `tbl`
Pemisahan Publik	File PHP terekspos di root aplikasi	Web root terisolasi di `/public` , file inti di `/app` aman dari akses publik
Pengujian (Testing)	Manual via browser	111 Automated Unit Tests (PHPUnit v10.5.16) dengan 428 assertions
Integrasi Social Share	`SocialShare.class.php` (AddThis, Twitter API, URL Shortener internal)	**Didepresiasi/Dihapus**. API pihak ketiga lama sudah usang; sharing di-handle native oleh frontend pembaca

1.2 Konvensi Penamaan Tabel & Penanganan DBPrefix

Pada CodeIgniter 4, prefix tabel dikonfigurasi secara global pada:

- `.env` : `database.default.DBPrefix = tbl`
- `app/Config/Database.php` : `'DBPrefix' => 'tbl'`

Aturan Eksekusi Query Builder:

1. **Penulisan di Kode CI4**: Seluruh pemanggilan model dan builder menggunakan **CamelCase tanpa awalan** `tbl` , misalnya:

```
$builder = $this->db->table('Article');
$builder->join('ArticleCategory ac', 'ac.articleID = a.articleID');
```

2. **Kompilasi Query SQL**: Query Builder secara otomatis meng-append prefix `'tbl'` , sehingga query yang dieksekusi ke database MySQL menjadi:

```
FROM tblArticle JOIN tblArticleCategory ac ON ac.articleID = a.articleID
```

3. **Peringatan Double Prefix**: Menuliskan `$db->table('tblArticle')` secara manual adalah **kesalahan**, karena CI4 akan mengeksekusi `tbltblArticle` yang menyebabkan error *table not found*.
4. **Paritas Linux Server**: Di server produksi berbasis Linux (yang bersifat *case-sensitive*), penulisan CamelCase seperti `'ArticleHeadline'` menjamin query merujuk persis ke tabel fisik `tblArticleHeadline` .

1.3 Peningkatan Keamanan & Autentikasi

1. **Formula Salt & Password Deterministik**:

```
$salt = hash('SHA512', '[' . strtolower($username) . '>|<' . $password . ']', false);
$hash = hash('SHA512', $salt . 'B3!21t454t03' . $password . $salt, false);
```

2. **Multi-Strategy Authentication & Auto-Upgrade**:
   - Sistem memvalidasi akun menggunakan formula SHA-512 Smarty di atas.
   - Jika akun lama terdeteksi masih menggunakan hash MD5 murni (`strlen(password) === 32`), sistem memvalidasi `md5($password) === $user['password']` .
   - Jika lolos, sistem **secara otomatis meng-upgrade hash akun ke format SHA-512 (128 karakter)** ke dalam database tanpa mengharuskan pengguna melakukan reset password manual.

3. **Autentikasi Dua Faktor (2FA) & Interval Verifikasi**:
   - Sistem login menerapkan autentikasi dua faktor berbasis kode verifikasi OTP 6-digit acak yang dikirimkan ke alamat email resmi staf redaksi via Brevo SMTP TLS.
   - Status kode OTP (`authCode`) dan masa aktifnya (`authExpired` , berlaku 5 menit) dikelola langsung pada entitas `tblUserAdministrator` . *(Catatan Keputusan Arsitektur: Smarty menetapkan `INTERVAL 60 SECOND` / 1 menit. Pada Revamp, masa berlaku diperpanjang menjadi 5 menit sebagai optimalisasi UX untuk mengakomodasi potensi latency pengiriman email gateway SMTP/Brevo tanpa mengorbankan keamanan).*
   - Setelah kode berhasil diverifikasi, sistem memberikan masa berlaku verifikasi selama 5 hari (`nextVerify = +5 days`) pada tabel pengguna. Dengan demikian, staf tidak diwajibkan menginput ulang kode verifikasi pada setiap sesi login harian selama periode tersebut masih aktif.

4. **Kebijakan Rotasi Sandi & Proteksi Akun**:
   - Batas masa aktif kata sandi staf ditetapkan selama 90 hari (`passwordExpired`). Filter rute `AuthGuard` secara otomatis mendeteksi akun yang telah melewati batas kedaluwarsa dan mengalihkannya ke halaman profil untuk pembaruan kata sandi.
   - Sesi pengguna diamankan menggunakan cookie bertanda tangan HMAC-SHA256 (`cmsinvestorid`) dengan atribut `HttpOnly` , `SameSite=Lax` , dan `Secure` pada server produksi.

5. **Sanitasi HTML Tingkat Lanjut (`HtmlSanitizer`)**:
   - Menggantikan sanitasi parsial `addslashes` pada Smarty.
   - Menghapus tag `<script>` , `<style>` , `<iframe>` tidak terpercaya, dan atribut inline event seperti `onclick` , `onload` , serta protokol pseudo `javascript:` .
   - Tetap mempertahankan tag format jurnalistik sah (`<p>` , `<strong>` , `<em>` , `<blockquote>` , `<a>` , `<img>` , `<table>`).

1.4 Spesifikasi Teknis Sistem Revamp CMS

Berikut adalah rincian spesifikasi teknis dan kebutuhan infrastruktur dari sistem Revamp CMS yang tengah dibangun:

Komponen	Spesifikasi & Rincian Versi
Bahasa & Runtime Engine	**PHP 8.3.28 (64-bit)** *(Kebutuhan versi minimum: PHP ^8.2)*. Fitur bahasa aktif: PSR-4 Autoloading, Strict Typing (`declare(strict_types=1)`), First-class callables, Enums, Readonly properties, dan JIT Compiler.

Komponen	Spesifikasi & Rincian Versi
Ekstensi PHP Wajib & Peran	• `ext-intl` : Internasionalisasi & lokalisasi format tanggal/waktu bahasa Indonesia. • `ext-mbstring` : Pemrosesan string multibyte UTF-8 untuk naskah berita, judul, dan slug. • `ext-mysqli` : Koneksi native MySQL 8 dengan dukungan query builder transaksional. • `ext-curl` : HTTP client untuk integrasi Brevo SMTP TLS, Cluster Media Storage, & Gemini API. • `ext-json` : Enkoding & dekoding payload JSON pada autosave draft, AJAX, dan REST API. • `ext-openssl` : Keamanan HMAC-SHA256 session token, hashing SHA-512, dan enkripsi payload. • `ext-gd` / `ext-imagick` : Manipulasi gambar lokal, pembuatan thumbnail, dan konversi WebP. • `ext-zip` : Pustaka kompresi dan ekstraksi berkas pada File Management.
Framework Fondasi	**CodeIgniter v4.7.4** (`codeigniter4/framework: v4.7.4` via Composer AppStarter). Komponen arsitektur: MVC terstruktur, Service Container (`Config\Services`), Route Filters / Middleware Guards (`AuthGuard` , `CsrfFilter` , `CorsFilter`), dan Spark CLI Engine.
Sistem Pengujian Mutu (Testing)	• **Unit Testing:** PHPUnit v10.5.16 (`phpunit/phpunit: ^10.5.16`) dengan 111 Test Cases dan 428 Assertions. • **Mocking Fixtures:** `mikey179/vfsStream: ^1.6` (Virtual Filesystem) & `fakerphp/faker: ^1.9` . • **End-to-End UI Testing:** Node.js v22.15.0 dengan Playwright Chromium Test Suite.
Web Server & Gateway	Nginx 1.22+ / Apache 2.4+ dengan PHP-FPM FastCGI (`pm = dynamic` , `max_children = 50` , `request_terminate_timeout = 120s`).
Struktur Web Root	Direktori *DocumentRoot* server diarahkan eksklusif ke folder `/public` . Seluruh file aplikasi inti (`/app`), pustaka (`/system`), dan berkas konfigurasi (`.env`) terisolasi dari akses web langsung.
Basis Data Engine	MySQL 8.0 / Percona Server 5.7+, Collation: `utf8mb4_general_ci` / `utf8mb4_unicode_ci` . Mendukung tabel InnoDB (transaksional) dan MyISAM legacy.
Penamaan Tabel (DBPrefix)	Prefix global `tbl` (`database.default.DBPrefix = tbl`). Model memanggil nama CamelCase tanpa prefix (`Article` , `Category`).
Antarmuka Redaksional (UI)	Native PHP Views terstruktur, Tailwind CSS, Lucide Icons, dan komponen JavaScript modern tanpa ketergantungan jQuery berat pada halaman baru.
Editor Teks Redaksi	CKEditor 4 tersanitasi dua lapis (browser-side & server-side), mendukung penataan paragraf, blockquote aman, embed video, dan pembersih spasi ganda.
Visualisasi Data	Highcharts & Chart.js 4 untuk render tabel finansial interaktif, template grafik (bar, line, pie), dan upload custom data CSV.
Penyimpanan Media & Berkas	Arsitektur Dual-Mode: 1. *Penyimpanan Intranet Produksi:* Integrasi REST API cluster media internal (`10.0.4.16` & `172.20.6.110`) dengan URL CDN publik `https://img.beritasatu.com/` . 2. *Fallback Lokal:* Folder terproteksi `/public/uploads/` dengan konvensi path `YYYY/MM/timestamp_name.webp` .
Email Gateway 2FA	Brevo SMTP TLS (Port 587) untuk pengiriman notifikasi kode verifikasi login staf redaksi.
Otomatisasi Background	Task runner berbasis CodeIgniter Spark CLI yang dieksekusi melalui Linux Crontab (penerbitan artikel terjadwal, truncate newsletter, sinkronisasi data saham IDX, dan pemrosesan antrean push notifikasi FCM).

2. Catatan Kritis: Status UI Preview Artikel

> **PERHATIAN KHUSUS TIM REDAKSI & FRONTEND INTEGRATOR:**
>
> Halaman pratinjau artikel pada rute `/post/preview/{id}` saat ini masih berupa **wrapper shell CMS internal** untuk memverifikasi kelengkapan relasi konten, metadata, thumbnail, dan pemformatan teks. Halaman ini **belum mengimplementasikan stylesheet dan komponen UI otentik dari portal publik investor.id**.
>
> **Tindakan yang harus dilakukan sebelum peluncuran penuh:**
>
> 1. Ekstrak layout styling artikel dari frontend publik portal `investor.id` (font family, typography lead paragraph, quote block styling, penataan gambar utama beserta caption, serta box saham terkait).
> 2. Perbarui view app/Views/post/preview.php menggunakan CSS otentik tersebut agar redaktur melihat tampilan yang 100% presisi dengan apa yang dilihat pembaca di situs live.

3. Komparasi Menyeluruh Seluruh Modul & Query Database (Smarty vs Revamp)

3.1 Modul Artikel & Redaksional (Editorial Listing)

- **Tabel Terkait**: `tblArticle`, `tblArticleCategory`, `tblCategory`, `tblUserAdministrator`, `tblMediaArticle`, `tblMedia`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/Article.class.php` $\rightarrow$ fungsi `Article::read()`
  - `Backup_CMS_ID/administration/pages/article/ajax/ajax_articlelist.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/ArticleModel.php $\rightarrow$ fungsi `getArticlesWithRelations()`
  - InvestorCMS/app/Controllers/Post.php $\rightarrow$ fungsi `index()`

Query di Smarty CMS:

```
SELECT a.*, c.categoryName, u.username, m.image, m.caption
FROM tblArticle a
LEFT JOIN tblArticleCategory ac ON ac.articleID = a.articleID
LEFT JOIN tblCategory c ON c.categoryID = ac.categoryID
LEFT JOIN tblUserAdministrator u ON u.userID = a.userID
LEFT JOIN tblMediaArticle ma ON ma.articleID = a.articleID
LEFT JOIN tblMedia m ON m.mediaID = ma.mediaID
WHERE a.status != 'Deleted'
ORDER BY a.articleDate DESC
LIMIT 0, 25;
```

Query di Revamp CMS:

```
$builder = $this->db->table('Article a')
    ->select('
        a.articleID,
        a.headline,
        a.slug,
        a.articleDate,
        a.status,
        a.viewed,
        a.source,
        a.editlock,
        u.name as author_name,
        u.username as author_username,
        c.categoryName as primary_category,
        c.categoryID as primary_category_id,
        m.image as thumbnail,
        m.caption as thumbnail_caption,
        lock_user.name as locked_by_name
    ')
    ->join('UserAdministrator u', 'u.userID = a.userID', 'left')
    ->join('UserAdministrator lock_user', 'lock_user.userID = a.editlock', 'left')
    ->join('ArticleCategory ac', 'ac.articleID = a.articleID', 'left')
    ->join('Category c', 'c.categoryID = ac.categoryID', 'left')
    ->join('MediaArticle ma', 'ma.articleID = a.articleID AND ma.position = 0', 'left')
    ->join('Media m', 'm.mediaID = ma.mediaID', 'left')
    ->where('a.status !=', 'Deleted')
    ->groupBy('a.articleID')
    ->orderBy('a.articleDate', 'DESC');
```

Optimasi: Thumbnail di Revamp dikunci secara eksplisit pada `ma.position = 0` guna mencegah duplikasi baris akibat multi-gambar, serta ditambahkan relasi ke `lock_user` untuk status kuncian editor.

3.2 Modul Detail Artikel, Master Kanal & Multi Sub-Kanal

- **Tabel Terkait**: `tblArticle`, `tblArticleCategory`, `tblCategory`, `tblArticleMeta`, `tblArticleAuthor`, `tblAuthors`, `tblArticleKeyword`, `tblKeyword`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/Article.class.php` $\rightarrow$ fungsi `Article::edit()`, `Article::getAuthor()`, `Article::getTags()`
  - `Backup_CMS_ID/administration/pages/article/ajax/ajax_articledetail.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/ArticleModel.php $\rightarrow$ fungsi `getArticleDetail()`
  - InvestorCMS/app/Controllers/Post.php $\rightarrow$ fungsi `edit()`

Pola di Smarty CMS:

Query dijalankan secara prosedural terpisah berkali-kali:

```
SELECT * FROM tblArticle WHERE articleID = 100;
SELECT * FROM tblArticleCategory WHERE articleID = 100;
SELECT * FROM tblArticleAuthor WHERE articleID = 100;
SELECT * FROM tblArticleKeyword WHERE articleID = 100;
```

Query di Revamp CMS:

```
// Ekstraksi data utama dan metadata dalam satu eksekusi
$article = $this->db->table('Article n')
    ->select('
        n.*,
        u.name as author_name,
        u.username as author_username,
        m.image as thumbnail,
        m.caption as thumbnail_caption,
        nm_prem.metavalue as is_premium,
        nm_deck.metavalue as upper_deck,
        nm_source.metavalue as news_source
    ')
    ->join('UserAdministrator u', 'u.userID = n.userID', 'left')
    ->join('MediaArticle ma', 'ma.articleID = n.articleID AND ma.position = 0', 'left')
    ->join('Media m', 'm.mediaID = ma.mediaID', 'left')
    ->join('ArticleMeta nm_prem', 'nm_prem.articleID = n.articleID AND nm_prem.metakey = "premium"', 'left')
    ->join('ArticleMeta nm_deck', 'nm_deck.articleID = n.articleID AND nm_deck.metakey = "upper_deck"', 'left')
    ->join('ArticleMeta nm_source', 'nm_source.articleID = n.articleID AND nm_source.metakey = "news_source"', 'left')
    ->where('n.articleID', $id)
    ->where('n.status !=', 'Deleted')
    ->get()->getRowArray();

// Ekstraksi seluruh kanal terikat (Master & Sub-Kanal)
$linkedCategories = $this->db->table('ArticleCategory ac')
    ->select('c.categoryID, c.categoryName, c.label, c.parent')
    ->join('Category c', 'c.categoryID = ac.categoryID')
    ->where('ac.articleID', $id)
    ->get()->getResultArray();
```

3.3 Modul Penulis, Editor & Fotografer (Authors)

- **Tabel Terkait**: `tblAuthors`, `tblArticleAuthor`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/Author.class.php` $\rightarrow$ fungsi `saveAuthor()`, `getAuthor()`, `searchAuthor()`
  - `Backup_CMS_ID/administration/pages/article/ajax/ajax_getauthor.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/AuthorModel.php $\rightarrow$ fungsi `getAuthorsWithCount()`, `findOrCreate()`

Query di Smarty CMS:

```
-- Smarty pencarian autocomplete penulis:
SELECT authorID, authorName FROM tblAuthors
WHERE authorName LIKE '%budi%' AND valid != 3
ORDER BY authorName LIMIT 0, 10;

-- Smarty hitung jumlah artikel penulis:
SELECT COUNT(authorID) FROM tblArticleAuthor WHERE authorID = 25;
```

Query di Revamp CMS:

```
public function getAuthorsWithCount(?string $keyword = null, int $limit = 20, int $offset = 0): array
{
    $builder = $this->db->table('Authors a')
        ->select('a.authorID, a.authorName, a.slug, a.type, COUNT(aa.articleID) as total_articles')
        ->join('ArticleAuthor aa', 'aa.authorID = a.authorID', 'left')
        ->where('a.valid !=', '3')
        ->groupBy('a.authorID')
        ->orderBy('a.authorName', 'ASC');

    if (!empty($keyword)) {
        $builder->like('a.authorName', $keyword);
    }

    return $builder->limit($limit, $offset)->get()->getResultArray();
}
```

3.4 Modul Taksonomi Kata Kunci & Tag (Keywords)

- **Tabel Terkait**: `tblKeyword`, `tblArticleKeyword`, `tblTag`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/Tag.class.php` $\rightarrow$ fungsi `getTagList()`, `saveTag()`, `getKeyword()`
  - `Backup_CMS_ID/administration/pages/article/ajax/ajax_taglist.php`
  - `Backup_CMS_ID/administration/pages/article/ajax/ajax_keywordlist.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/KeywordModel.php $\rightarrow$ fungsi `syncArticleKeywords()`, `getArticleKeywords()`

Query di Smarty CMS:

```
SELECT SQL_CALC_FOUND_ROWS tagID, label, slug FROM tblTag
WHERE label LIKE 'saham%' ORDER BY label ASC LIMIT 0, 10;
```

Query di Revamp CMS:

```
public function syncArticleKeywords(int $articleId, array $keywords): void
{
    // Bersihkan relasi lama
    $this->db->table('ArticleKeyword')->where('articleID', $articleId)->delete();

    foreach ($keywords as $kw) {
        $cleanKw = trim($kw);
        if ($cleanKw === '') continue;

        // Cari atau buat master keyword baru
        $existing = $this->db->table('Keyword')->where('keyword', $cleanKw)->get()->getRowArray();
        if ($existing) {
            $keywordId = $existing['keywordID'];
        } else {
            $slug = url_title($cleanKw, '-', true);
            $this->db->table('Keyword')->insert(['keyword' => $cleanKw, 'slug' => $slug]);
            $keywordId = $this->db->insertID();
        }

        // Tautkan pivot
        $this->db->table('ArticleKeyword')->insert([
            'articleID' => $articleId,
            'keywordID' => $keywordId
        ]);
    }
}
```

3.5 Modul Slot Headline Terkurasi & Sorting

- **Tabel Terkait**: `tblArticleHeadline`, `tblArticleHeadlineHistory`, `tblArticle`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/Article.class.php` (Headline handlers)
  - `Backup_CMS_ID/administration/pages/article/headline.php`
  - `Backup_CMS_ID/administration/pages/article/ajax/ajax_headlinelist.php`
  - `Backup_CMS_ID/administration/pages/article/ajax/ajax_headlinesave.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/HeadlineModel.php $\rightarrow$ fungsi `getActiveHeadlines()`, `batchReorder()`
  - InvestorCMS/app/Controllers/Headline.php
  - InvestorCMS/app/Commands/TruncateNewsletterHeadline.php

Query di Smarty CMS:

```
SELECT h.*, a.headline, a.articleDate
FROM tblArticleHeadline h
INNER JOIN tblArticle a ON a.articleID = h.articleID
WHERE h.category = 'home' AND h.status = 1
ORDER BY h.num ASC;
```

Query di Revamp CMS:

```
public function batchReorder(string $category, array $orderedIds, string $user = 'admin'): bool
{
    $this->db->transStart();
    foreach ($orderedIds as $position => $headlineId) {
        $this->db->table('ArticleHeadline')
            ->where('id', $headlineId)
            ->where('category', $category)
            ->update([
                'num'       => $position + 1,
                'set_by'    => $user,
                'timestamp' => date('Y-m-d H:i:s'),
            ]);

        // Audit log otomatis ke ArticleHeadlineHistory
        $this->db->table('ArticleHeadlineHistory')->insert([
            'articleID' => $headlineId,
            'category'  => $category,
            'action'    => 'REORDER',
            'position'  => $position + 1,
            'user'      => $user,
            'notes'     => 'Urutan headline diperbarui via drag & drop editor',
            'timestamp' => date('Y-m-d H:i:s')
        ]);
    }
    $this->db->transComplete();
    return $this->db->transStatus();
}
```

3.6 Modul Media & Manajemen Gambar

- **Tabel Terkait**: `tblMedia`, `tblMediaArticle`, `tblAlbum`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/Media.class.php` $\rightarrow$ fungsi `getMediaList()`, `totalMedia()`
  - `Backup_CMS_ID/administration/pages/media/index.php`
  - `Backup_CMS_ID/administration/pages/media/ajax/ajax_medialist.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/MediaModel.php $\rightarrow$ fungsi `getMediaWithRelations()`, `getArticlesUsingMedia()`
  - InvestorCMS/app/Controllers/Media.php

Query di Smarty CMS:

```
SELECT SQL_CALC_FOUND_ROWS t1.mediaID, t1.caption, t1.description, t1.filename, t1.timestamp, t1.mediaTitle, t1.source, t1.author, t2
FROM tblMedia AS t1, tblUserAdministrator AS t2
WHERE t1.userID = t2.userID AND t1.filename != ''
ORDER BY t1.mediaID DESC LIMIT 0, 20;
```

Query di Revamp CMS:

```
public function getMediaWithRelations(?string $search = null, int $page = 1, int $perPage = 20): array
{
    $builder = $this->db->table('Media m')
        ->select('m.*, u.username as uploader_name, COUNT(ma.articleID) as total_used')
        ->join('UserAdministrator u', 'u.userID = m.userID', 'left')
        ->join('MediaArticle ma', 'ma.mediaID = m.mediaID', 'left')
        ->where('m.filename !=', '')
        ->groupBy('m.mediaID')
        ->orderBy('m.mediaID', 'DESC');

    if (!empty($search)) {
        $builder->groupStart()
            ->like('m.caption', $search)
            ->orLike('m.mediaTitle', $search)
            ->groupEnd();
    }

    return $builder->limit($perPage, ($page - 1) * $perPage)->get()->getResultArray();
}
```

Arsitektur Penyimpanan Media Dual-Driver (`MediaStorageService`)

Revamp CMS mengisolasi mekanisme penyimpanan berkas gambar menggunakan service terpadu InvestorCMS/app/Libraries/MediaStorageService.php. Layanan ini mendukung dua mode operasional yang ditentukan melalui variabel environment `MEDIA_STORAGE_DRIVER`:

1. **Lingkungan Development / Lokal (`MEDIA_STORAGE_DRIVER = local`)**:
   - File fisik disimpan langsung di direktori web root lokal: `public/uploads/media/YYYY/MM/{timestamp}_{prefix}_{hash}.jpg`.
   - Gambar diproses dan dikompresi menggunakan GD Library lokal dengan batas resolusi maksimum 1200px dan kualitas JPEG 85%.
   - Kolom database `tblMedia.filename` tetap menyimpan format standar legacy: `YYYY/MM/namafile.jpg`.
   - URL publik lokal disajikan melalui rute web: `http://localhost:8080/uploads/media/YYYY/MM/namafile.jpg`.
2. **Lingkungan Produksi (`MEDIA_STORAGE_DRIVER = smarty_api` / `legacy_api`)**:
   - Gambar tidak disimpan di server web lokal CMS, melainkan di-stream melalui cURL POST dengan payload Base64 ke klaster microservice upload internal AWS S3 (`http://10.0.4.16/imgapi/upload/`, dengan server cadangan di `http://172.20.6.110:8080/imgapi/upload/`).
   - Microservice tersebut menyimpan file fisik ke storage server terpusat `/data/media/investor/YYYY/MM/` dan secara otomatis menghasilkan varian format WebP serta thumbnail multi-resolusi (`620x350`, `320x220`, `120x120`).
   - Konten gambar didistribusikan ke publik pembaca melalui CDN BeritaSatu (`https://img.beritasatu.com/` dan `https://img2.beritasatu.com/investor/`).
   - Jika koneksi ke microservice gagal/unreachable, sistem secara otomatis *fallback* menyimpan sementara ke storage lokal untuk mencegah naskah gagal terbit.

Standarisasi Pemanggilan Gambar: Helper Global `media_url()`

Untuk memastikan tampilan gambar di seluruh view redaksi, editor, dan reader API tidak rusak terlepas dari apakah sistem berjalan di lokal atau produksi, Revamp CMS menyediakan helper global InvestorCMS/app/Helpers/media_helper.php dengan fungsi `media_url(?string $filename)`:

```
// Contoh pemanggilan di view Blade / PHP:
<img src="<?= media_url($article['thumbnail']) ?>" alt="Thumbnail">
```

Fungsi `media_url()` bekerja secara cerdas menangani 4 varian format nilai database:

1. **Full URL Eksternal**: Jika bernilai `https://...` atau `http://...` (misal gambar dari CDN AWS S3 langsung atau scraping berlisensi), nilai URL dikembalikan langsung tanpa modifikasi.
2. **Format Path Lokal** `/uploads/...` : Otomatis diselesaikan menggunakan fungsi `base_url(ltrim($filename, '/'))` .
3. **Format Standar Legacy** `YYYY/MM/namafile.jpg` :
   - Pada mode `local` : Memeriksa keberadaan file fisik di `public/uploads/media/` . Jika ada, menghasilkan `http://localhost:8080/uploads/media/YYYY/MM/namafile.jpg` .
   - Pada mode `production` : Otomatis menggabungkan nilai dengan `MEDIA_CDN_BASE_URL` sehingga menghasilkan `https://img.beritasatu.com/YYYY/MM/namafile.jpg` .
4. **Fallback File Kosong**: Jika kolom bernilai `null` atau string kosong, fungsi otomatis mengembalikan URL placeholder default: `base_url('img/investor_placeholder.jpg')` .

3.7 Modul Pengguna Administrator & RBAC

- **Tabel Terkait**: `tblUserAdministrator`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/Administrator.class.php` $\rightarrow$ fungsi `auth()` , `encodePassword()` , `getAdminList()`
  - `Backup_CMS_ID/administration/pages/user/index.php`
  - `Backup_CMS_ID/administration/pages/user/add.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/AdminModel.php $\rightarrow$ fungsi `authenticate()` , `getAdminsWithActivity()`
  - InvestorCMS/app/Controllers/User.php

Query di Smarty CMS:

```
SELECT * FROM tblUserAdministrator
WHERE username = 'radyan' AND active = 1 LIMIT 0, 1;
```

Query di Revamp CMS:

```
public function getAdminsWithActivity(): array
{
    return $this->db->table('UserAdministrator u')
        ->select('
            u.userID,
            u.username,
            u.name,
            u.email,
            u.roleName,
            u.active,
            MAX(l.timestamp) as last_active_log,
            COUNT(DISTINCT a.articleID) as total_articles_created
        ')
        ->join('EditorLog l', 'l.userID = u.userID', 'left')
        ->join('Article a', 'a.userID = u.userID', 'left')
        ->groupBy('u.userID')
        ->orderBy('u.userID', 'ASC')
        ->get()->getResultArray();
}
```

Mekanisme Custom Permission per User (Granular Access Override)

Selain RBAC berbasis peran standar (`master` , `administrator` , `editor` , `reporter` , dll), Revamp CMS menyediakan fitur *Custom Permission* berbasis file JSON (`writable/data/json/permissions.json`) yang dievaluasi langsung oleh filter RoleGuard.php.

Fitur ini memungkinkan Administrator memberikan atau membatasi izin akses spesifik modul untuk user tertentu (misal: membatasi reporter hanya bisa mengakses modul `post` dan `media` , atau memberi staf akses ke `embed-chart`). Format konfigurasi JSON:

```
{
  "users": {
    "12": {
      "dashboard": true,
      "post": true,
      "media": true,
      "headline": false,
      "files": false
    }
  }
}
```

Pengelolaan izin ini dapat diakses melalui controller InvestorCMS/app/Controllers/User.php via metode `getPermissions()` dan `savePermissions()` .

3.8 Modul Penguncian Artikel Bersamaan (Edit Lock)

- **Tabel Terkait**: `tblArticle` , `tblUserAdministrator`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/Article.class.php` $\rightarrow$ fungsi `Article::PostLock()` , `Article::ReleaseLock()`
  - `Backup_CMS_ID/administration/pages/article/ajax/ajax_releaselock.php`
- **Lokasi File Revamp**:

- InvestorCMS/app/Models/ArticleModel.php $\rightarrow$ fungsi `acquireLock()`, `releaseLock()`, `forceUnlock()`

Mekanisme Query di Revamp CMS:

```
// Acquire Lock
$row = $this->db->table('Article n')
    ->select('n.editlock, u.name as locked_by_name, u.username as locked_by_username')
    ->join('UserAdministrator u', 'u.userID = n.editlock', 'left')
    ->where('n.articleID', $articleId)
    ->get()->getRowArray();

if ($row['editlock'] === null || (int)$row['editlock'] === $userId) {
    $this->db->table('Article')
        ->where('articleID', $articleId)
        ->update(['editlock' => $userId]);
    return ['status' => true, 'locked' => false];
}

// Release Lock
$this->db->table('Article')
    ->where('articleID', $articleId)
    ->where('editlock', $userId)
    ->update(['editlock' => null]);
```

3.9 Modul Riwayat Revisi & Audit Trail (Article History)

- **Tabel Terkait**: `tblArticleHistory`, `tblArticle`, `tblUserAdministrator`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/Article.class.php` $\rightarrow$ fungsi `Article::Revision()`
  - `Backup_CMS_ID/administration/pages/article/ajax/ajax_articlehistory.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/ArticleModel.php $\rightarrow$ fungsi `getRevisionHistory()`, `getRevisionDetail()`

Query di Revamp CMS:

```
public function getRevisionHistory(int $page = 1, int $perPage = 25, array $filters = []): array
{
    $builder = $this->db->table('ArticleHistory h')
        ->select('
            h.id,
            h.timestamp,
            h.articleID,
            h.headline,
            h.editorID,
            u.name as editor_name,
            u.username as editor_username,
            h.action,
            h.status_prev,
            h.status_after
        ')
        ->join('Article a', 'a.articleID = h.articleID', 'left')
        ->join('UserAdministrator u', 'u.userID = h.editorID', 'left')
        ->orderBy('h.timestamp', 'DESC');

    if (!empty($filters['article_id'])) {
        $builder->where('h.articleID', (int)$filters['article_id']);
    }

    return $builder->limit($perPage, ($page - 1) * $perPage)->get()->getResultArray();
}
```

3.10 Modul Visualisasi Grafik & Embed Data (ChartData)

- **Tabel Terkait**: `tblChartData`, `tblUserAdministrator`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/administration/pages/files/chart.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/ChartModel.php
  - InvestorCMS/app/Controllers/EmbedChart.php

Query di Smarty CMS:

```
SELECT SQL_CALC_FOUND_ROWS t1.chartID AS id, t1.chartTitle AS title, t1.createdAt, t2.username, t1.source
FROM tblChartData AS t1, tblUserAdministrator AS t2
WHERE deletedAt IS NULL AND t1.createdBy = t2.userID
ORDER BY t1.chartID DESC;
```

Query di Revamp CMS:

```
public function getChartsList(?string $search = null, int $limit = 20, int $offset = 0): array
{
    $builder = $this->db->table('ChartData c')
        ->select('c.chartID, c.chartTitle, c.chartType, c.dataJson, c.source, c.createdAt, u.username as creator_name')
        ->join('UserAdministrator u', 'u.userID = c.createdBy', 'left')
        ->where('c.deletedAt IS NULL', null, false)
        ->orderBy('c.chartID', 'DESC');

    if (!empty($search)) {
        $builder->like('c.chartTitle', $search);
    }

    return $builder->limit($limit, $offset)->get()->getResultArray();
}
```

3.11 Modul Arsip Dokumen & Integrasi Google Docs (Files)

- **Tabel Terkait**: `tblFiles`, `tblUserAdministrator`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/administration/pages/files/index.php`
  - `Backup_CMS_ID/administration/pages/files/googledocs.php`
  - `Backup_CMS_ID/administration/pages/files/api_googledocs.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/FileModel.php
  - InvestorCMS/app/Models/GoogleDocsModel.php
  - InvestorCMS/app/Controllers/Files.php

Query di Smarty CMS:

```
SELECT SQL_CALC_FOUND_ROWS t1.fileID, t1.title, t1.filename, t1.timestamp, t2.username
FROM tblFiles AS t1, tblUserAdministrator AS t2
WHERE t1.userID = t2.userID ORDER BY t1.title ASC;
```

Query di Revamp CMS:

```
public function getFilesWithUploader(?string $keyword = null, int $limit = 20, int $offset = 0): array
{
    $builder = $this->db->table('Files f')
        ->select('f.fileID, f.title, f.filename, f.filetype, f.filesize, f.timestamp, u.username as uploader_name')
        ->join('UserAdministrator u', 'u.userID = f.userID', 'left')
        ->orderBy('f.timestamp', 'DESC');

    if (!empty($keyword)) {
        $builder->like('f.title', $keyword);
    }

    return $builder->limit($limit, $offset)->get()->getResultArray();
}
```

3.12 Modul Notifikasi Push & Ekosistem Aplikasi Mobile

- **Tabel Terkait**: `tblPushNotif`, `tblPushToDevices`, `tblPushRegistered`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/PushNotif.php`
  - `Backup_CMS_ID/administration/pages/push/index.php`
  - `Backup_CMS_ID/administration/pages/push/save.php`
  - `Backup_CMS_ID/administration/pages/misc/apps.php`
  - `Backup_CMS_ID/cron/pushnotif.php` & `Backup_CMS_ID/cron/pushfcm.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/NotificationModel.php
  - InvestorCMS/app/Controllers/Apps.php
  - InvestorCMS/app/Commands/ProcessPushNotif.php

Alur Eksekusi Push Notification:

1. **Redaksi Mengirim Pesan**: Disimpan ke `tblPushNotif` dengan status `'ready'`.
2. **Cron Scheduler CLI**: Perintah `php spark cron:process-push` memproses antrean:

```
$pending = $db->table('PushNotif')->where('status', 'ready')->get()->getResultArray();
foreach ($pending as $item) {
    // Kirim ke Google FCM HTTP v1 API
    $db->table('PushNotif')->where('id', $item['id'])->update(['status' => 'sent']);
}
```

3.13 Modul Statistik, Trafik & Laporan Produktivitas

- **Tabel Terkait**: `tblArticle`, `tblUserAdministrator`, `tblAuthors`, `tblCategory`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/lib/Report.class.php` $\rightarrow$ fungsi `getProductivity()`
  - `Backup_CMS_ID/administration/pages/statistic/index.php`
  - `Backup_CMS_ID/administration/pages/statistic/export.php`

- **Lokasi File Revamp**:
  - InvestorCMS/app/Controllers/Statistic.php
  - InvestorCMS/app/Controllers/StatisticReport.php
  - InvestorCMS/app/Models/ArticleModel.php $\rightarrow$ fungsi `getTrafficChartData()`, `getPublishingChartData()`

Query Agregasi Produktivitas Editor di Smarty:

```
SELECT tmp.editor, count(tmp.articleID) as ttl_article, sum(tmp.views) as ttl_views
FROM (
    SELECT a.articleid, a.userid, a.headline, a.articledate, a.viewed as views, e.name as editor
    FROM tblArticle a
    INNER JOIN tblUserAdministrator e ON a.userid = e.userid
    WHERE a.status = 'publish' AND a.articledate BETWEEN '2026-10-01 00:00:00' AND '2026-10-20 23:59:59'
) tmp
GROUP BY tmp.editor ORDER BY tmp.editor ASC;
```

Query Heatmap & Hourly Traffic di Revamp CMS:

```
$todayRows = $this->db->table('Article')
    ->select('HOUR(articleDate) as h, SUM(COALESCE(viewed, 0)) as total_v')
    ->where('DATE(articleDate)', $todayStr)
    ->groupBy('HOUR(articleDate)')
    ->get()->getResultArray();
```

3.14 Modul Konten Fitur (Promotional Banners & Tags)

- **Tabel Terkait**: `tblFeatureContent`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/administration/pages/feature/index.php`
  - `Backup_CMS_ID/administration/pages/feature/save.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/FeatureContentModel.php
  - InvestorCMS/app/Controllers/FeatureContent.php

Query di Smarty CMS:

```
SELECT * FROM tblFeatureContent WHERE id = '5' AND type = 'banner' AND position = 'top';
```

Query di Revamp CMS:

```
public function getFeaturesList(string $type = 'banner'): array
{
    return $this->db->table('FeatureContent')
        ->where('type', $type)
        ->orderBy('position', 'ASC')
        ->get()->getResultArray();
}
```

3.15 Modul Keanggotaan Pembaca (Member)

- **Tabel Terkait**: `tblMember`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Models/MemberModel.php
  - InvestorCMS/app/Controllers/Member.php

Query di Revamp CMS:

```
public function getMembers(array $filters = [], int $page = 1, int $perPage = 10): array
{
    $builder = $this->db->table('Member');
    if (!empty($filters['q'])) {
        $q = trim($filters['q']);
        $builder->groupStart()
            ->like('fullname', $q)
            ->orLike('email', $q)
            ->groupEnd();
    }
    if (!empty($filters['status'])) {
        $builder->where('status', $filters['status']);
    }
    return $builder->orderBy('signupdate', 'DESC')
        ->limit($perPage, ($page - 1) * $perPage)
        ->get()->getResultArray();
}
```

3.16 Modul Trending Topics & Peringkat Baca

- **Tabel Terkait**: `tblArticle`, `tblArticleKeyword`, `tblKeyword`
- **Lokasi File Smarty**:
  - `Backup_CMS_ID/administration/pages/article/trending.php`
  - `Backup_CMS_ID/administration/pages/article/ajax/ajax_trendingtopic.php`
- **Lokasi File Revamp**:
  - InvestorCMS/app/Controllers/Trending.php
  - InvestorCMS/app/Models/ArticleModel.php $\rightarrow$ fungsi `getTopArticles()`

Query di Revamp CMS:

```
public function getTopArticles(int $limit = 10, string $interval = '7d'): array
{
    $days = ($interval === '24h') ? 1 : (($interval === '30d') ? 30 : 7);
    $since = date('Y-m-d H:i:s', strtotime("-{$days} days"));

    return $this->db->table('Article a')
        ->select('a.articleID, a.headline, a.slug, a.viewed, a.articleDate, c.categoryName')
        ->join('ArticleCategory ac', 'ac.articleID = a.articleID', 'left')
        ->join('Category c', 'c.categoryID = ac.categoryID', 'left')
        ->where('a.status', 'Publish')
        ->where('a.articleDate >=', $since)
        ->groupBy('a.articleID')
        ->orderBy('a.viewed', 'DESC')
        ->limit($limit)
        ->get()->getResultArray();
}
```

3.17 Modul Background Scheduler & Cron Jobs

Tugas Cron	File Smarty	File Revamp	Pola Query & Aksi
Auto-Publish Scheduled	`Backup_CMS_ID/cron/updatedraft.php`	app/Commands/PublishScheduled.php	`UPDATE tblArticle SET status = 'Publish' WHERE status = 'Hold' AND articleDate <= NOW()`
Truncate Newsletter Headline	`Backup_CMS_ID/cron/truncate_newsletter.php`	app/Commands/TruncateNewsletterHeadline.php	`UPDATE tblArticleHeadline SET status = 0 WHERE category = 'newsletter'` (pukul 08:00, 14:00, 20:00 WIB)
Proses Antrean Push FCM	`Backup_CMS_ID/cron/pushnotif.php`	app/Commands/ProcessPushNotif.php	`SELECT FROM tblPushNotif WHERE status = 'ready'`, dispatch FCM v1
Pembersihan Log Audit Lama	`Backup_CMS_ID/cron/deletelog.php`	app/Commands/CleanupLogs.php	`DELETE FROM tblEditorLog WHERE timestamp < NOW() - INTERVAL 90 DAY`
Feed Data Saham & Indeks	`Backup_CMS_ID/cron/index-saham.php`	app/Commands/FetchIdxStock.php	Parsing XML/FTP feed IDX, update cache ticker

3.18 Modul Evaluasi & Kesiapan SEO (SEO Readiness Scoring)

- **Tabel Terkait**: `tblSeoScores` (skema DDL di §5.2)
- **Status di Smarty CMS**: ❌ Tidak ada (Fitur baru pada Revamp CMS)
- **Lokasi File Revamp**:
  - Controller: InvestorCMS/app/Controllers/Scoring.php
  - Library Evaluator: InvestorCMS/app/Libraries/SeoScorer.php
  - Model: InvestorCMS/app/Models/SeoScoreModel.php
  - View: `app/Views/scoring/index.php`

Logika Evaluasi Algoritma SEO:

Modul ini menghitung skor kesiapan naskah (0 - 100) sebelum diterbitkan redaksi berdasarkan bobot parameter berikut:

1. **Title Tag Optimization (20%)**: Panjang judul ideal (50-65 karakter), ketiadaan duplikasi, dan penempatan kata kunci utama.
2. **Focus Keyword Presence & Density (25%)**: Keberadaan keyword di paragraf pembuka (lead), kepadatan kata kunci optimal (1.5% - 2.5%), serta kemunculan pada sub-judul.
3. **Content Length & Readability (25%)**: Minimal panjang naskah editorial berita (300+ kata) dan keterbacaan paragraf.
4. **Heading Structure (15%)**: Keberadaan hierarki sub-heading H2 dan H3 dalam naskah.
5. **Meta Description & Media Caption (15%)**: Kelengkapan ringkasan naskah (summary) dan caption gambar penjelas.

Histori penilaian disimpan ke dalam tabel `tblSeoScores` untuk memantau peningkatan kualitas artikel sebelum dipublikasikan. Modul ini juga terhubung dengan layanan Google Gemini AI (`GeminiProvider.php`) untuk memberikan saran penyempurnaan judul dan paragraf pembuka secara instan.

---

4. Spesifikasi & Pemanggilan Seluruh Endpoint API

4.1 Reader Public APIs

1. List Artikel Publik

- **Endpoint**: `GET /api/articles`
- **Controller**: InvestorCMS/app/Controllers/Api/ArticleApi.php
- **Parameters**: `page` (default: 1), `per_page` (default: 10, max: 50), `category` (slug/ID), `q` (kata kunci)
- **Sample Output**:


```
{
  "status": "success",
  "data": [
    {
      "articleID": 982145,
      "headline": "IHSG Ditutup Menguat Menjelang Keputusan Suku Bunga BI",
      "slug": "ihsg-ditutup-menguat-menjelang-keputusan-suku-bunga-bi",
      "articleDate": "2026-10-20 16:00:00",
      "primary_category": "Market",
      "author_name": "Raka Herdika",
      "thumbnail": "http://localhost:8080/uploads/media/2026/10/ihsg.jpg",
      "viewed": 1420
    }
  ],
  "pagination": { "current_page": 1, "per_page": 10, "total_rows": 145020, "total_pages": 14502 }
}
```


2. Detail Lengkap Artikel

- **Endpoint**: `GET /api/articles/(:num)`
- **Sample Output**:


```
{
  "status": "success",
  "data": {
    "articleID": 982145,
    "headline": "IHSG Ditutup Menguat Menjelang Keputusan Suku Bunga BI",
    "summary": "Indeks Harga Saham Gabungan melaju di zona hijau...",
    "body": "<p>Jakarta, Investor.id - Indeks Harga Saham Gabungan...</p>",
    "articleDate": "2026-10-20 16:00:00",
    "categories": [{ "categoryID": 10, "categoryName": "Market", "label": "market" }],
    "authors": [{ "authorID": 45, "authorName": "Raka Herdika", "type": "reporter" }],
    "tags": ["IHSG", "Bursa Efek Indonesia", "Suku Bunga"],
    "is_premium": 0
  }
}
```

4.2 Market Data & Bursa APIs

- **Market Ticker**: `GET /api/market/ticker` $\rightarrow$ Indeks gabungan (IHSG, LQ45, IDX30), nilai tukar mata uang asing, harga komoditas (Minyak WTI, Emas, CPO).
- **Market Movers**: `GET /api/market/movers?type=gainers` $\rightarrow$ 10 saham top gainers, losers, dan most active hari ini.

4.3 Editorial & Newsroom APIs

1. Autosave Draft Artikel (Asinkron)

- **Endpoint**: `POST /api/post/autosave`
- **Controller**: InvestorCMS/app/Controllers/Api/PostApi.php
- **Request Body (JSON)**:

```
{
  "articleID": 982145,
  "headline": "IHSG Ditutup Menguat Menjelang Keputusan Suku Bunga BI",
  "summary": "Ringkasan draf...",
  "body": "<p>Isi naskah berita...</p>",
  "categoryID": 10
}
```

2. AI Editorial Assistant (Gemini API)

- **Endpoint**: `POST /api/ai/generate`
- **Request Body**:

```
{
  "action": "suggest_titles",
  "content": "Isi lengkap naskah berita...",
  "tone": "formal"
}
```

- **Actions yang Didukung**: `suggest_titles` (3 opsi judul SEO-friendly), `generate_summary` (ringkasan 2 paragraf), `proofread` (koreksi EYD/PUEBI), `extract_tags` (rekomendasi tag relevan).

3. Autocomplete Tag & Keyword Redaksi

- **Endpoint**: `POST /post/search-tags` (Payload: `{ "term": "saham" }`)

4. SEO Readiness Scoring & AI Recommendations

- **Endpoint Analisis**: `POST /api/scoring/analyze`
  - **Payload**: `{ "article_id": 982145 }` atau `{ "headline": "...", "body": "...", "keywords": ["saham", "ihsg"] }`
  - **Response**: Breakdown skor parameter SEO (Title, Keyword, Body Length, Headings, Meta) beserta saran perbaikan.
- **Endpoint Histori**: `GET /api/scoring/history/(:num)`
  - Mengambil daftar histori penilaian artikel berdasarkan ID artikel.

4.4 Reader Authentication APIs

- **Client Login**: `POST /api/auth/client-login` (Payload: `email`, `password`)
- **Client Register**: `POST /api/auth/client-register` (Payload: `name`, `email`, `password`)

---

5. Hal-Hal yang Wajib Dilakukan Sebelum Naik ke Produksi (Pre-Production Checklist & Runbook)

Sebelum sistem Revamp CMS dialihkan menjadi sistem produksi utama (*cutover*), tim teknis (Sysadmin, DevOps, dan Software Engineer) wajib menjalankan serangkaian tahapan persiapan dan validasi berikut secara berurutan:

5.1 Verifikasi Konfigurasi Environment (`.env`)

Pastikan seluruh parameter pada file `.env` di server produksi telah disesuaikan dengan lingkungan operasional live:

```
# 1. Mode Operasional Aplikasi
CI_ENVIRONMENT = production

# 2. Base URL Resmi Portal CMS
app.baseURL = 'https://cms.investor.id/'

# 3. Konfigurasi Koneksi Database Produksi (Wajib DBPrefix = tbl)
database.default.hostname = 127.0.0.1
database.default.database = investor
database.default.username = prod_user
database.default.password = [PROD_PASSWORD]
database.default.DBDriver = MySQLi
database.default.DBPrefix = tbl
database.default.DBDebug = false
database.default.charset  = utf8mb4
database.default.DBCollat = utf8mb4_general_ci

> **Catatan Konsistensi Collation**: Database eksisting Smarty dan server database produksi menggunakan `utf8mb4_general_ci`. Pastika

# 4. Kredensial Email Gateway 2FA (Brevo SMTP TLS)
email.SMTPHost = 'smtp-relay.brevo.com'
email.SMTPPort = 587
email.SMTPCrypto = 'tls'
email.SMTPUser = '[BREVO_SMTP_USER]'
email.SMTPPass = '[BREVO_SMTP_PASSWORD]'
email.fromEmail = 'noreply@b-universe.id'
email.fromName = 'CMS Investor.ID'

# 5. Media Storage Driver (Cluster Intranet Produksi)
MEDIA_STORAGE_DRIVER = smarty_api
MEDIA_API_ENDPOINT = 'http://10.0.4.16/imgapi/upload/'
MEDIA_SECONDARY_API_ENDPOINT = 'http://172.20.6.110:8080/imgapi/upload/'
MEDIA_CDN_BASE_URL = 'https://img.beritasatu.com/'
MEDIA_CDN_ORIGINAL_URL = 'https://img2.beritasatu.com/investor/'

# 6. Integrasi AI Editorial (Gemini API)
GEMINI_API_KEY = [LIVE_GEMINI_KEY]
```

5.2 Eksekusi Skema Database, Tabel Baru & Indexing Kinerja

Jalankan perintah migrasi atau eksekusi DDL berikut pada database `investor` untuk memastikan tabel-tabel pendukung baru dan indeks query kinerja tinggi telah terpasang:

```
-- 1. Tabel Sub-headline & Poin Penting Artikel
CREATE TABLE IF NOT EXISTS `tblArticlePoints` (
  `id` INT(11) UNSIGNED NOT NULL AUTO_INCREMENT,
  `articleID` INT(11) NOT NULL,
  `point_order` INT(11) NOT NULL DEFAULT 1,
  `point_text` TEXT NOT NULL,
  `createdAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  KEY `idx_articleID` (`articleID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- 2. Tabel Audit Trail Perubahan Posisi Headline
CREATE TABLE IF NOT EXISTS `tblArticleHeadlineHistory` (
  `id` INT(11) UNSIGNED NOT NULL AUTO_INCREMENT,
  `articleID` INT(11) NOT NULL,
  `category` VARCHAR(50) NOT NULL DEFAULT 'home',
  `action` ENUM('ADD','REMOVE','REORDER','TRUNCATE') NOT NULL,
  `position` INT(11) DEFAULT NULL,
  `user` VARCHAR(100) NOT NULL DEFAULT 'SYSTEM',
  `notes` VARCHAR(255) DEFAULT NULL,
  `timestamp` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  KEY `idx_history_articleID` (`articleID`),
  KEY `idx_history_category` (`category`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- 3. Tabel Skoring SEO Editorial
CREATE TABLE IF NOT EXISTS `tblSeoScores` (
  `id` INT(11) UNSIGNED NOT NULL AUTO_INCREMENT,
  `articleID` INT(11) NOT NULL,
  `score` INT(3) NOT NULL DEFAULT 0,
  `keyword` VARCHAR(255) DEFAULT NULL,
  `density` DECIMAL(5,2) NOT NULL DEFAULT 0.00,
  `recommendations` TEXT DEFAULT NULL,
  `updatedAt` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  UNIQUE KEY `uniq_article_seo` (`articleID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- 4. Tabel Audit Log Asisten AI
CREATE TABLE IF NOT EXISTS `tblAiGeneratedLogs` (
  `id` INT(11) UNSIGNED NOT NULL AUTO_INCREMENT,
  `userID` INT(11) NOT NULL,
  `articleID` INT(11) DEFAULT NULL,
  `action` VARCHAR(50) NOT NULL,
  `prompt_tokens` INT(11) DEFAULT 0,
  `completion_tokens` INT(11) DEFAULT 0,
  `model_used` VARCHAR(50) NOT NULL,
  `timestamp` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (`id`),
  KEY `idx_ai_user` (`userID`),
  KEY `idx_ai_article` (`articleID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

-- 5. Indexing Kinerja untuk Tabel Utama Berkapasitas Jutaan Baris
```

```
ALTER TABLE `tblArticle` ADD INDEX IF NOT EXISTS `idx_status_date` (`status`, `articleDate`);
ALTER TABLE `tblArticleCategory` ADD INDEX IF NOT EXISTS `idx_art_cat` (`articleID`, `categoryID`);
ALTER TABLE `tblMediaArticle` ADD INDEX IF NOT EXISTS `idx_media_art_pos` (`articleID`, `position`);
ALTER TABLE `tblArticleHeadline` ADD INDEX IF NOT EXISTS `idx_cat_status_num` (`category`, `status`, `num`);
ALTER TABLE `tblArticleKeyword` ADD INDEX IF NOT EXISTS `idx_kw_art` (`keywordID`, `articleID`);
```

Alternatif eksekusi via Spark CLI:

```
php spark migrate
```

5.3 Konfigurasi Web Server & Isolasi Direktori Publik

Arahkan *DocumentRoot* virtual host Nginx atau Apache ke sub-direktori `/public`. File-file sensitif seperti `.env`, folder `/app`, `/system`, dan `/writable` sama sekali tidak boleh terekspos ke internet.

Contoh Blok Konfigurasi Nginx (`/etc/nginx/sites-available/investorcms.conf`):

```
server {
    listen 80;
    server_name cms.investor.id;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl http2;
    server_name cms.investor.id;

    # SSL Certificate
    ssl_certificate /etc/letsencrypt/live/cms.investor.id/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/cms.investor.id/privkey.pem;

    # Document Root wajib mengarah ke folder /public
    root /var/www/investorcms/public;
    index index.php index.html;

    client_max_body_size 32M;

    # Routing CodeIgniter 4
    location / {
        try_files $uri $uri/ /index.php$is_args$args;
    }

    # Blokir akses langsung ke berkas tersembunyi
    location ~ /\. {
        deny all;
        access_log off;
        log_not_found off;
    }

    # PHP-FPM FastCGI
    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/run/php/php8.3-fpm.sock; # Atau /run/php/php8.2-fpm.sock (sesuai versi PHP aktif)
        fastcgi_param SCRIPT_FILENAME $realpath_root$fastcgi_script_name;
        include fastcgi_params;
    }
}
```

5.4 Perizinan Hak Akses Folder Server (`writable/`)

Aplikasi CodeIgniter 4 membutuhkan hak akses tulis pada direktori kerja untuk penyimpanan file cache, sesi, file log, dan folder unggahan lokal:

```
# Menetapkan ownership ke user web server
chown -R www-data:www-data /var/www/investorcms

# Memberikan akses tulis ke direktori kerja aplikasi
chmod -R 775 /var/www/investorcms/writable
chmod -R 775 /var/www/investorcms/public/uploads

# Memastikan direktori cache dan logs siap digunakan
mkdir -p /var/www/investorcms/writable/cache
mkdir -p /var/www/investorcms/writable/logs
mkdir -p /var/www/investorcms/writable/session
```

5.5 Penjadwalan Crontab Otomatisasi Linux

Buka editor crontab pengguna web server (`crontab -e -u www-data`) dan pasang tugas otomatisasi berikut:

```
# 1. Terbitkan naskah terjadwal yang telah melewati jadwal tayang (tiap 1 menit)
* * * * * cd /var/www/investorcms && php spark cron:publish-scheduled > /dev/null 2>&1

# 2. Kosongkan / truncate slot headline newsletter (pukul 08:00, 14:00, 20:00 WIB)
0 8,14,20 * * * cd /var/www/investorcms && php spark cron:truncate-newsletter > /dev/null 2>&1

# 3. Tarik pergerakan data bursa saham IDX (tiap 10 menit pada jam perdagangan Senin-Jumat)
*/10 9-16 * * 1-5 cd /var/www/investorcms && php spark cron:fetch-idx-stock > /dev/null 2>&1

# 4. Proses antrean siaran push notification FCM (tiap 1 menit)
* * * * * cd /var/www/investorcms && php spark cron:process-push-notif > /dev/null 2>&1

# 5. Pembersihan log audit editor berusia lebih dari 90 hari (tiap hari pukul 02:00 WIB)
0 2 * * * cd /var/www/investorcms && php spark cron:cleanup-logs > /dev/null 2>&1
```

5.6 Pengujian Pre-Flight Operasional (Smoke Test Checklist)

Sebelum merilis URL CMS ke seluruh staf redaksi, lakukan verifikasi manual pada poin-poin berikut:

- ☐ **Uji Login & 2FA**: Lakukan login menggunakan akun uji coba. Pastikan email OTP 6-digit masuk ke inbox dalam < 10 detik, input kode verifikasi, dan pastikan dashboard terbuka normal.
- ☐ **Uji Sesi 5 Hari**: Logout dan login kembali dengan akun yang sama dalam hari yang sama. Pastikan sistem langsung masuk tanpa meminta OTP ulang (validasi grace period 5 hari).
- ☐ **Uji Pembuatan Artikel**: Buat draft naskah berita baru lengkap dengan judul, slug auto-generate, kanal, sub-kanal, keyword chips, dan cover foto. Simpan naskah dan pastikan muncul di daftar artikel.
- ☐ **Uji Pessimistic Lock**: Buka naskah yang sama dari dua browser berbeda dengan akun berbeda. Pastikan browser kedua menampilkan peringatan bahwa naskah sedang dikunci oleh redaktur pertama.
- ☐ **Uji Kurasi Headline**: Masukkan naskah ke slot Headline kategori 'home' dan lakukan drag-and-drop urutan nomor 1 s.d. 5. Pastikan perubahan nomor tersimpan di database.
- ☐ **Uji Upload Media & Reverse Lookup**: Unggah foto baru ke Media Library, gunakan sebagai cover naskah, lalu coba hapus foto tersebut dari Media Library. Pastikan sistem memblokir penghapusan karena foto sedang aktif dipakai artikel.
- ☐ **Uji Ekspor Laporan**: Buka menu Laporan Editorial dan ekspor publish log ke format CSV. Buka di spreadsheet dan pastikan karakter serta inisial alias editor terbaca rapi.

5.7 Rencana Kontingensi & Prosedur Rollback

Jika pada hari peluncuran ditemukan kendala kritis yang tidak dapat diselesaikan dalam window maintenance (misal: gangguan koneksi database cluster atau kendala jaringan intranet):

1. **Pengalihan DNS / Virtual Host**:
   - Kembalikan konfigurasi web server Nginx/Apache untuk mengarahkan domain `cms.investor.id` ke direktori sistem lama Smarty CMS (`/var/www/backup_cms_id/administration/pages/login/index.php`).
   - Reload web server: `sudo systemctl reload nginx` .
2. **Integritas Basis Data**:
   - Karena Revamp CMS menggunakan skema nama tabel fisik asli (`tblArticle` , `tblArticleAuthor` , `tblCategory` , `tblUserAdministrator`), data naskah baru yang sempat dibuat di Revamp tetap tersimpan aman dan dapat langsung dibaca oleh sistem Smarty CMS lama tanpa perlu konversi ulang.
3. **Penyelidikan Masalah**:
   - Periksa file log error CodeIgniter pada direktori `/var/www/investorcms/writable/logs/log-YYYY-MM-DD.php` untuk menelusuri penyebab kegagalan teknis.